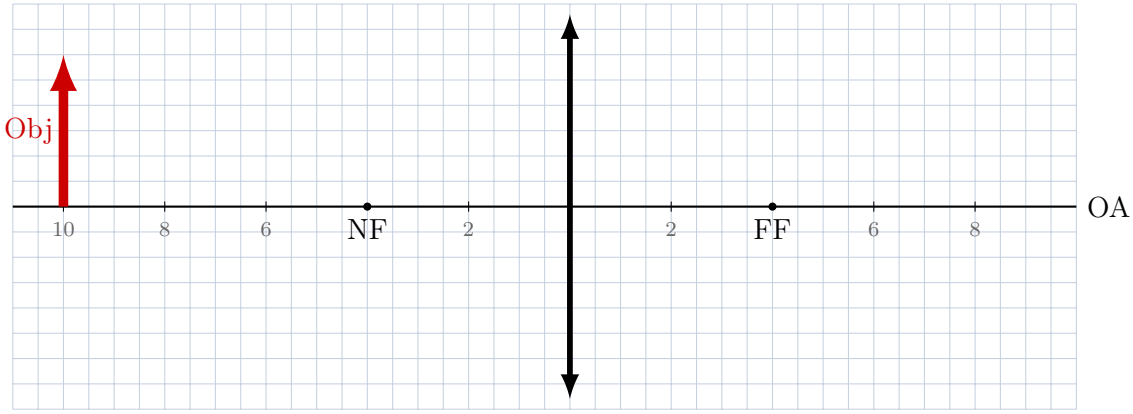
1. A converging lens has $f = 4$ cm. A 3 cm tall object stands 10 cm in front of it. On the grid, one small square is 0.5 cm.



A. Draw all **three** principal rays from the tip of the object. Mark the image.

B. From your drawing, estimate d_i, h_i and M.

$d_i \approx$ _____ $h_i \approx$ _____ $M \approx$ _____

C. Is the image real or virtual? Upright or inverted? Bigger or smaller?

2. An optometrist grinds a double convex lens from plastic with $n = 1.6$. The first surface has $|R_1| = 20$ cm and the second has $|R_2| = 30$ cm.

A. What are the **signs** of R_1 and R_2? Explain using "surface then C" or "C then surface."

B. Use the lens maker's equation to find f.

C. What is the power of this lens, in diopters? Is it converging or diverging?